Andrey Derzhavin

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Summary

System software engineer with 5+ years of experience at Huawei RII, focused on CPU functional simulation, microarchitectural design space exploration, and compiler optimisations. Skilled in C++, LLVM, and JIT compilation. Proven ability to lead research projects from prototype to product integration. Passionate about teaching and mentoring (co-created a university course). Looking for opportunities in compilers, emulation, or high-performance computing.

Education

Moscow Institute of Physics and Technology — *2023 – 2025 / Dolgoprudny, Russia*

M.S in applied math and physics, honours degree

Moscow Institute of Physics and Technology — *2019 – 2023 / Dolgoprudny, Russia*

B.S in applied math and physics, honours degree

Physics and math lyceum 30 — *2017 – 2019 / Saint Petersburg, Russia*

Extra Education

"Uses and applications of C++ language" — *Sep 2020 – Apr 2021*

MIPT course. Lecturer – K. Vladimirov, Intel.

Experience

Huawei RRI — *Jul 2021 – Present / Moscow, Russia*

- Co-developed and co-taught "CPU & OS Simulation" course (MIPT, ITMO)
- Led a functional simulation project from early prototype to successful integration into another product within a year (team of two).
- Conducted research on improving microarchitectural Design space exploration with low-order parameters. Developed a parallel, simulator-independent framework for finding optimal uArch configuration
- Implemented tool which finds loops in program's CFG using Boost Graph. Conducted research on dynamic cycles classification, estimated potential performance improvement from enhancing loop termination prediction using ISA hint

Acronis Internship — *Jul 2020 – Nov 2020 / Remote*

- Learned right-context grammars, proposed & investigated several ways of applying them to static analysis of C++ source code

Projects

leech-compiler — *Sep 2023 – May 2024*

Educational project from 5th course

- Implementation of compiler IR for custom VM. Developed several popular optimization passes & analyses

JitResearch — *Sep 2025 – Present*

Research project

- Conducted a comparative study of modern JIT libraries, evaluating performance, flexibility, and integration complexity for CPU simulation purposes.

Publications & Talks

JIT libraries for CPU simulation: difficulties of choice — *May 2026*

C++Russia 2026 talk

Skills

Programming Languages	C | C++ | Python
Software Development	git | make | CMake | Linux (WSL2, Ubuntu) | LaTeX
Frameworks & Libraries	LLVM IR | OpenCL | Numpy | Boost
Languages	***Russian:*** native ***English:*** C1(IELTS 7.0)

Extra

- Interested in compilers, system programming, functional simulation
- Hobbies – football, history